

Q1. Restriction enzymes are used during genetic engineering. The target sites for EcoRI, BamHI and AluI is shown in the table below.

EcoRI	5' ...G↓AATTC ...3' 3' ...CTTAA↑G ...5'
BamHI	5' ...G↓GATCC ...3' 3' ...CCTAG↑G ...5'
AluI	5' ...AG↓CT...3' 3' ...TC↑GA...5'

Table 1.1

A bacterial plasmid is shown in the figure below. There are no target sequences for the three restriction enzymes in the rest of the plasmid.

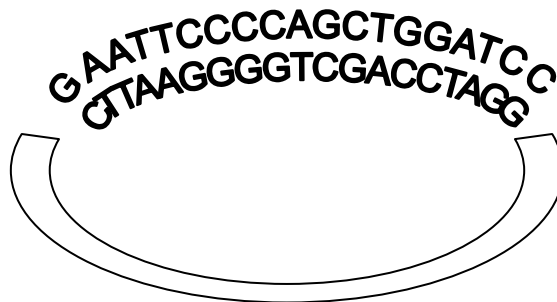


Fig 1.1

With reference to **Table 1.1** given above,

a) State how many fragments of DNA will be present after the plasmid has been cut with

i) a mixture of EcoRI and BamHI [1]

2 fragments;

ii) a mixture of EcoRI [1]

1 fragments;

b) Explain why it is more suitable to use EcoRI than AluI in genetic engineering of the plasmid.[2]

EcoRI results in sticky ends while AluI results in blunt ends;

Using AluI requires additional step using terminal transferase to add nucleotides to create overlap;

c) A student incubates the plasmid molecule shown in Figure 1.1 with EcoRI to insert a human gene.

Outline how restriction enzymes may be used in the formation of recombinant DNA molecules.[3]

Use same restriction enzyme to cut both plasmid and foreign gene;

Breaks/hydrolyses phosphodiester bond/ sugar-phosphate backbone;

Incubate plasmid and foreign gene together, anneal by CBP/H bonding;

Ligase added, catalyse formation of phosphodiester bonds;

d) Name two other enzymes used in the formation of recombinant DNA and briefly describe their roles.[2]

terminal transferase: catalyse addition of nucleotides to blunt ends;

reverse transcriptase: catalyse formation of DNA molecule from a RNA template;

ligase: catalyse formation of phosphodiester bonds;

e) Gel electrophoresis is usually performed after restriction digestion to separate DNA fragment. After which, nucleic acid hybridization is conducted.

i) Briefly describe how DNA fragments may be separated by gel electrophoresis.[3]

fragments separated based on size and charge;

DNA loaded in well at negative end of electrode, DNA is negatively charged and migrates to the positive electrode/ anode;

Rate of migration is inversely proportional to size of fragment / appropriate description of speed of fragments + relative position of fragment reference to well position;

ii) Describe how the process of nucleic acid hybridization is carried out. [4]

DNA in gel denatured to form ssDNA and transferred to membrane/blot;

Crosslinked to permanently link ssDNA to membrane;

ssDNA probes radioactively labeled, complementary to sequence of foreign gene added to membrane;

binds to DNA fragment containing foreign gene by CPB/ H bonding;

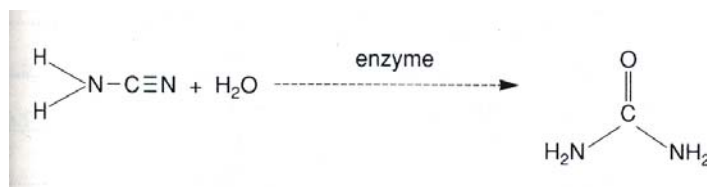
autoradiography / x-ray film exposed by radioactive probe;

black band shows position of DNA fragment which contains gene of interest/ foreign gene;

(TOTAL: 16 marks)

2. Read the following passage about genetic engineering and answer the questions that follow.

A 5% cyanamide kills the weeds which infest fields of tobacco plants, but causes some damage to the tobacco. However, tobacco plants have been genetically modified by inserting a gene from a soil fungus, *Myrothecium verrucaria*. The gene codes for an enzyme, cyanamide hydratase, which converts cyanamide into urea:



The gene was inserted into a plant virus and then tobacco plants were infected with the virus. A strain of transgenic tobacco plants was thus developed that carried the gene cyanamide hydratase but were unaffected by other viral proteins. The enzyme was most active in the leaves, showing about one-fifth of the activity found in *Myrothecium*.

The transgenic plants thrive when treated with a dilute solution of cyanamide, since the plants have urease enzymes which convert urea to ammonium ions which provide the plants with a source of nitrogen.

a) Explain why the tobacco plants described in the passage are transgenic plants. [1]

carry and express genes from other species/ genes not found in native plants/
possesses gene introduced from another organism, whether from same or different species;

b) Suggest why the enzymes activity in the transgenic plant is lower than the fungus from which the gene was taken. [2]

conditions in transgenic plants different from the fungus;

not optimal conditions results in lower rate of enzyme activity;

presence of ammonium ions may result in diff pH / temp;

c) Outline the possible benefits and potential hazards of this particular example of a genetically modified organism. [4]

Benefits

S: allows transgenic tobacco be sprayed with cyanamide which kills only weeds;

E: reduces competition for nutrients/ space/ light/ resources and increases crop yield;

S: transgenic plants convert cyanamide to ammonium ions;

E: function as fertilizer resulting in more robust plants/ cut costs for using lesser fertilizer;

Hazards

S: transgene may be transferred to wild-type tobacco/ related plants;
E: effect on transgene on other types of organisms which feed on tobacco/ related plants not known;
E: cyanamide may toxic to beneficial insects / adversely affect insectt biodiversity;

The use of viruses as a vector to deliver the cyanamide hydratase gene is one method of delivering the transgene to the plant. Another commonly used method is the use of soil bacterium *Agrobacterium tumeficiens* as a vector.

d) State two other methods which can be used to deliver foreign genes to plants. [1]

electroporation;
microprojectile;

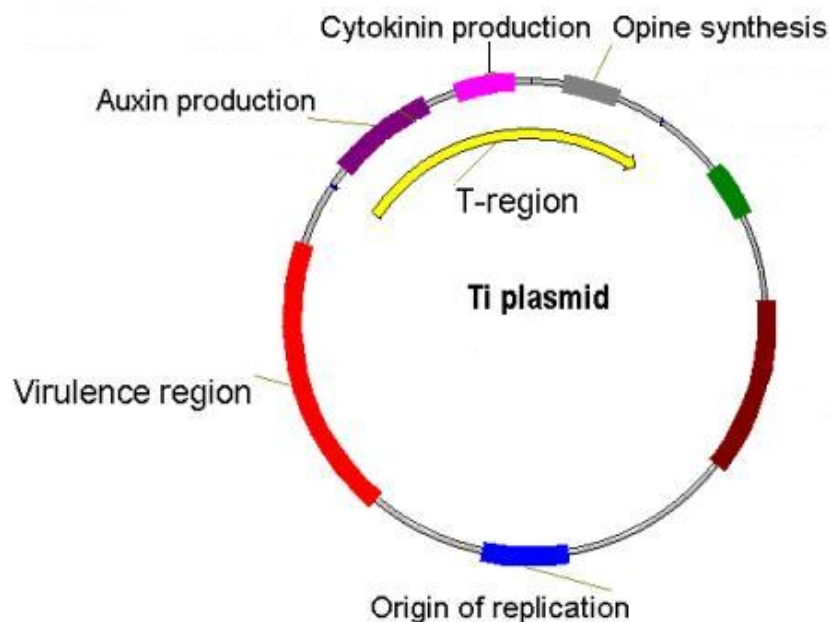


Fig 2.1

e) The Ti plasmid in *Agrobacterium tumeficiens* has to be attenuated and modified before it can be used as a vector for genetic engineering. With reference to **Fig 2.1**, explain what this means. [3]

auxin, cytokinin and opine stimulate plant cell growth;

plant hormones synthesis genes removed;

to prevent tumour formation;

insert a selection marker gene;

virulence region remains as it is needed for excision, transfer and integration of T-DNA region;

f) Describe how a genetically modified tobacco plant containing the cyanamide hydratase transgene can be produced using *Agrobacterium tumefaciens*-mediated leaf disc transformation. [4]

T-DNA and transgene cut with same RE and ligated together;

Recombinant T plasmids transformed into *A. tumefaciens*, used to infect leaf disc / recombinant plasmids introduced by electroporation into leaf disc;

Leaf disc placed on culture medium containing nutrients + grows into callus;;

Add auxin triggers root formation + cytokinin triggers shoot formation;

By adjusting conc of A:C, cells are induced to differentiate into plantlet;

Plantlet transplanted to soil/peat for further growth;

Q3. Every day when she wakes up, Lisa begins to deal with a condition she's known all her life — **cystic fibrosis**. Her parents clap her back and chest for at least 20 minutes to help clear her lungs of the thick mucus that sometimes makes it difficult for her to breathe.

At school, Lisa coughs a lot, so she keeps a box of tissues on her desk just in case she coughs up mucus and needs to spit it out. In gym class, she participates in sports but often gets tired easily. At lunchtime, Lisa takes pills to help her digest her food and get all the vitamins she needs, which helps her keep up her strength to deal with cystic fibrosis.

(a) Explain what is the cause of cystic fibrosis [2]

Gene mutation of CFTR protein;

Heritable, pass from parent to offspring;

(b) Describe a suitable therapy that can help Lisa in leading more or less a normal lifestyle. [4]

Gene therapy;

Functional copy of CFTR gene carried by vector/ adenovirus;

Attenuated virus infects target cells;

Delivery via aerosol which reaches respiratory tissues;

Synthesis of functional CFTR protein which is inserted into plasma membrane of target cell;

(c) Explain the problems of the suggested therapy and suggest why Lisa will have to repeat the treatment periodically. [4]

Functional CFTR gene not integrated in genome, not stable and degraded after some time;

S: Short-lived nature;

E: functional copy of gene not integrated/ cells have short lifespan & die/ immune system kills virus vector;

S: may trigger cancer;

E: may be inserted within protooncogenes or tumour suppressor genes/ may be inserted into control elements which affect gene expression of genes

S: Immune response triggered by viral vector;

E: repeated treatments may result in adverse effects due to immune response/ toxicity of viral vectors;

S: cannot treat multigene disorders, different sites;

E: gene therapy only targets respiratory tissues but other diseased sites such as pancreas & reproductive organs are not treated;

Genetically-modified fish, which can grow up to 10 times faster than normal, could be cleared for human consumption within a year. A US firm - AF Protein - is developing the GM fish on Prince Edward Island, Canada.

The fish would become the first genetically modified animal cleared for consumption, and the technology involved could cut the cost of raising salmon and trout by half. The Massachusetts-based company has inserted two sets of fish genes into Atlantic salmon. The first set are growth hormone genes and the second, from a different fish, activate them.

As a result, at the age of 18 months, the salmon are five times the size of their unmodified siblings.

Article taken from BBC News 18 May 1999

Discuss the impact of genetic modified salmon on fishermen in Canada. [4]

Transgenic fish may be more aggressive, when escaped, compete against other wild type fish. May result in decrease in variety of fish;

Market competition as it is cheaper than ocean caught fish/ may not be commercially viable to fish;

Potential diseases in not yet known in transgenic fish may decrease yield of fish;

Total: 14

FRQ

- (i) Discuss the benefits and ethical concerns of the Human Genome Project
[10]

1. provides a detailed physical, genetic map of the human genome and DNA sequence of entire human genome
2. allow elucidation of genes associated with human diseases/ understanding genetics basis of disease;
3. allows early detection or diagnosis via genetic testing;
4. e.g risk of breast cancer, diabetes and colon cancer;
5. improve diagnosis diseases prenatally;
6. pharmacogenomics : design customized drugs based on genetic profile;
7. enables novel treatment through gene therapy as gene sequences are now readily available;
8. pharmaceutical products such as medicine can be synthesized using readily available gene sequences; e.g. humulin and tissue plasminogen activator (TPA) given to after heart attack to dissolve blood clots;
9. a means of risk assessment to evaluate health risks of individuals who may be exposed to radiation or carcinogens;

max : 6 points

10. ethical issues of pre-implantation genetic testing viewed as “tampering with nature” by certain religious groups;
11. To-be parents may have to make difficult decisions of terminating pregnancy due to presence of genetic disorder. It may amount to murder since there are different definitions of when a foetus is recognized as living;
12. positive test result indicates that a person possesses a known mutation and may not necessarily develop cancer. Are the general population informed adequately;
13. management of genetic information, who owns the right to the genetic information? The person sample taken from or the hospital or company who sequenced the information?
14. work-related discrimination: is it right for employers to demand DNA testing because it can reveal any possible high risk disease the employee may develop and thus increase costs of employing this person;
15. insurance-related discrimination: is it right for insurance companies to demand DNA testing because it can reveal any possible high risk disease which is not covered under the insurance policy;
16. Designer drugs developed by pharmacogenomics discriminates against the poor as only the wealthy have the ability to pay for such services;

Max: 4 points

(ii) Explain the uses of RFLP analysis.[6]

1. Genomic mapping;
2. RFLP of a specific length is used as a genetic marker;
3. used to find out how RFLPs are linked/ to make linkage/ physical map for genome;

Max : 2 points

4. Disease detection / screening for diseased alleles;
5. Presence of RFLP closely linked to diseased alleles indicate
6. useful for screening parents' DNA for genetic counseling/ screen foetus's DNA to detect disease;

Max : 2 points

7. DNA fingerprinting;
8. used in forensics to identity criminals / used in paternity determination/ determination of bodies during diasters /

(iii) Explain the properties, using named examples of the normal functions of stem cells. [4]

1. named type of stem cells: Embryonic, Adult, cord blood stem cells;
2. unspecialized;
3. self-renewal;
4. give rise to specialized cell type;
5. potency of stem cells;